Day 111 | Unit 08 | Teacher plan

Solve Area, Surface Area, and Volume Problems

Build and Interpret Nets

Standards: 6.G.A.4

Learning target: I can draw a net and account for every face of a solid.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. A net unfolds all faces into a plane without overlap. Rectangular prisms have three matching pairs of rectangles. Pyramids have a polygonal base and triangular sides. Add face areas, not edge lengths.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 4

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01 / Learn

A net unfolds all faces into a plane without overlap. Rectangular prisms have three matching pairs of rectangles. Pyramids have a polygonal base and triangular sides. Add face areas, not edge lengths.

Worked example 1

Draw a net for a closed 7 × 6 × 2 cm rectangular prism. Label the six faces and calculate total area.

1. A net unfolds all faces into a plane without overlap.

2. Rectangular prisms have three matching pairs of rectangles.

3. Add 2(42) + 2(14) + 2(12) = 136.

4. State the result in the form requested: 2 faces each of 7×6, 7×2, 6×2; 136 cm².

Answer: 2 faces each of 7×6, 7×2, 6×2; 136 cm²

Worked example 2

A closed rectangular prism is 7 cm by 2 cm by 2 cm. Select all true descriptions of its net.
A. Two faces are 7 by 2.
B. Its net contains exactly four triangles.
C. Two faces are 7 by 2 and two are 2 by 2.
D. Adding just the three distinct face areas counts every face.

1. A net unfolds all faces into a plane without overlap.

2. A closed rectangular prism has 6 rectangular faces in 3 matching pairs.

3. Count both copies of each face.

Answer: A, C - Two faces are 7 by 2.; Two faces are 7 by 2 and two are 2 by 2.

Day 111 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a net for a closed 4 × 4 × 4 cm rectangular prism. Label the six faces and calculate total area.

02 / Constructed response

Draw a net for a closed 8 × 3 × 2 cm rectangular prism. Label the six faces and calculate total area.

03 / Constructed response

Draw a net for a closed 7 × 5 × 2 cm rectangular prism. Label the six faces and calculate total area.

04 / Constructed response

Draw a net for a closed 3 × 7 × 5 cm rectangular prism. Label the six faces and calculate total area.

05 / Constructed response

Draw a net for a closed 8 × 7 × 4 cm rectangular prism. Label the six faces and calculate total area.

06 / Constructed response

Draw a net for a closed 2 × 4 × 3 cm rectangular prism. Label the six faces and calculate total area.

07 / Constructed response

Draw a net for a triangular prism. Its two congruent right-triangle bases have legs 3 cm and 4 cm and hypotenuse 5 cm; prism length is 4 cm. Label all faces and find surface area.

08 / Multiple select

A closed rectangular prism is 6 cm by 4 cm by 4 cm. Select all true descriptions of its net.
A. Two faces are 6 by 4.
B. Its net contains exactly four triangles.
C. Two faces are 6 by 4 and two are 4 by 4.
D. Adding just the three distinct face areas counts every face.

09 / Multiple select

A closed rectangular prism is 6 cm by 5 cm by 3 cm. Select all true descriptions of its net.
A. Its net contains exactly four triangles.
B. Adding just the three distinct face areas counts every face.
C. Two faces are 6 by 5.
D. Two faces are 6 by 3 and two are 5 by 3.

10 / Constructed response

Draw a net for a triangular prism. Its two congruent right-triangle bases have legs 3 cm and 4 cm and hypotenuse 5 cm; prism length is 6 cm. Label all faces and find surface area.

Day 111 | Unit 08 | Answer key

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1. 2 faces each of 4×4, 4×4, 4×4; 96 cm²

Add 2(16) + 2(16) + 2(16) = 96.

6.G.A.4 | Constructed response

2. 2 faces each of 8×3, 8×2, 3×2; 92 cm²

Add 2(24) + 2(16) + 2(6) = 92.

6.G.A.4 | Constructed response

3. 2 faces each of 7×5, 7×2, 5×2; 118 cm²

Add 2(35) + 2(14) + 2(10) = 118.

6.G.A.4 | Constructed response

4. 2 faces each of 3×7, 3×5, 7×5; 142 cm²

Add 2(21) + 2(15) + 2(35) = 142.

6.G.A.4 | Constructed response

5. 2 faces each of 8×7, 8×4, 7×4; 232 cm²

Add 2(56) + 2(32) + 2(28) = 232.

6.G.A.4 | Constructed response

6. 2 faces each of 2×4, 2×3, 4×3; 52 cm²

Add 2(8) + 2(6) + 2(12) = 52.

6.G.A.4 | Constructed response

7. 2 triangles (area 6 each), rectangles 3×4, 4×4, 5×4; SA 60 cm²

Triangle pair: 12 cm²; rectangles: (3+4+5) × 4 = 48; total 60.

6.G.A.4 | Constructed response

8. A, C - Two faces are 6 by 4.; Two faces are 6 by 4 and two are 4 by 4.

A closed rectangular prism has 6 rectangular faces in 3 matching pairs. Count both copies of each face.

6.G.A.4 | Multiple select

9. C, D - Two faces are 6 by 5.; Two faces are 6 by 3 and two are 5 by 3.

A closed rectangular prism has 6 rectangular faces in 3 matching pairs. Count both copies of each face.

6.G.A.4 | Multiple select

10. 2 triangles (area 6 each), rectangles 3×6, 4×6, 5×6; SA 84 cm²

Triangle pair: 12 cm²; rectangles: (3+4+5) × 6 = 72; total 84.

6.G.A.4 | Constructed response